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Zero-adjusted defective regression models for modeling lifetime data

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ABSTRACT

In this paper, we introduce a defective regression model for survival data modeling with a proportion of early failures or zero-adjusted. Our approach enables us to accommodate three types of units, that is, patients with 'zero' survival times (early failures) and those who are susceptible or not susceptible to the event of interest. Defective distributions are obtained from standard distributions by changing the domain of the parameters of the latter in such a way that their survival functions are limited to $p \in (0, 1)$. We consider the Gompertz and inverse Gaussian defective distributions, which allow modeling of data containing a cure fraction. Parameter estimation is performed by maximum likelihood estimation, and Monte Carlo simulation studies are conducted to evaluate the performance of the proposed models. We illustrate the practical relevance of the proposed models on two real data sets. The first is from a study of occlusion of endoscopic stenting in patients with pancreatic cancer performed at A.C. Camargo Cancer Center, and the other is from a study on insulin use in pregnant women diagnosed with gestational diabetes performed at São Paulo University Medical School. Both studies were performed in São Paulo, Brazil.

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1. Introduction

In medical studies, it is common that some units under study are not susceptible to the event of interest, called immune or cured elements. A class of models, referred to as cure rate models, considers these situations and has been studied by several authors in the recent years. The most popular model is the standard mixture model, which was first proposed by Boag [5] and Berkson and Gage [4]. In this model, the population survival function is $S(t) = p_1 + (1 - p_1)S_0(t)$, such that $p_1 \in (0, 1)$ is referred to as the cured fraction, and $S_0(t)$ is a proper survival function for uncured patients. Standard mixture model is widely

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